Muscle Memory for Machines: A Spatiotemporal Motor Cache for Autonomous GUI Agents

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***Abstract*—Current AI computer use agents interact with graphical user interfaces statelessly—every click is treated as a first-time click, with no benefit from prior interactions. This contrasts sharply with human motor behavior, where repeated actions become faster and more automatic through practice. We present a spatiotemporal motor memory cache that enables autonomous GUI agents to develop "muscle memory." The system integrates eight complementary mechanisms: R-Tree spatial indexing, Power Law of Practice speed-up, Fitts's Law movement timing, WindMouse trajectory adaptation, temporal decay, EMA position smoothing, action sequence prediction, and cross-session persistence. A three-layer resolution hierarchy—model-provided hints, accessibility API detection, and percentage-bucketed window-relative coordinates—ensures that *every* mouse interaction builds cache entries, regardless of application or platform. Testing on real desktop applications across four operating systems demonstrates monotonically growing cache hit counts across sessions and across independent agent instances, with movement durations converging to 30% of initial values for frequently-used targets. To our knowledge, this is the first system to integrate cognitive science motor learning models into the action-execution layer of AI GUI agents.**

***Index Terms*—computer use agents, muscle memory, motor learning, Fitts's Law, Power Law of Practice, spatial caching, GUI automation**

I. Introduction

A human expert's 100th click on "Save" is fast and automatic—a ballistic movement requiring no conscious attention. An AI computer use agent's 100th click on the same button takes exactly as long as its first. This asymmetry reveals a fundamental gap in current autonomous GUI systems: they have no concept of motor learning.

Recent advances in large language models (LLMs) have enabled a new class of *computer use agents* (CUAs) that interact with desktop applications through screenshots and mouse/keyboard input [1], [2], [3]. These systems have demonstrated impressive reasoning capabilities—deciding *what* to click based on visual understanding. However, the execution layer—*how* to reach the target—remains entirely stateless. Every mouse movement is computed from scratch, with no benefit from prior interactions with the same UI elements.

This stands in stark contrast to human motor behavior. Decades of cognitive science research have established that repeated aimed movements follow the *Power Law of Practice* [4]: movement time decreases as a power function of repetition count. Additionally, the base timing of aimed movements obeys *Fitts's Law* [5], and unused motor memories decay following patterns consistent with the *Ebbinghaus forgetting curve* [6].

We present a *spatiotemporal motor memory cache* that bridges this gap. Our system sits entirely in the execution layer, orthogonal to and composable with any LLM backbone. It integrates eight complementary mechanisms:

1) **R-Tree spatial indexing** for proximity-based element lookup
2) **Power Law of Practice** speed-up ($T(n) = T(1) \cdot n^{-0.4}$)
3) **Fitts's Law** base movement timing
4) **WindMouse trajectory adaptation** (gravity, wind, velocity)
5) **Temporal decay** (7-day half-life) for cache freshness
6) **EMA position smoothing** (α=0.3) for element drift
7) **Sequence prediction** for anticipatory targeting
8) **Cross-session SQLite persistence**

A three-layer resolution hierarchy ensures universal coverage: model-provided semantic hints (Layer 3), accessibility API auto-detection (Layer 2), and percentage-bucketed window-relative coordinates (Layer 1). Layer 1 requires only the foreground window's bounding rectangle—available on every major OS—ensuring that *every* click builds muscle memory, even in applications with no accessibility support.

II. Background and Related Work

A. Cognitive Science Foundations

Power Law of Practice. Newell and Rosenbloom [4] established that the time to perform a task decreases as a power function of the number of practice trials: $T(n) = T(1) \cdot n^{-\alpha}$, where α typically ranges from 0.2 to 0.6 for motor tasks. This law has been observed across diverse domains including cigar rolling, card sorting, and typing.

Fitts's Law. Fitts [5] showed that the time required for an aimed movement is logarithmically related to the ratio of distance to target width: $MT = a + b \cdot \log_2(D/W + 1)$. This remains the fundamental model of aimed movements in HCI [7], [8].

Ebbinghaus Forgetting Curve. Ebbinghaus [6] demonstrated that memory strength decays exponentially with time. We implement this as a half-life model where cache entry confidence halves every 7 days.

Motor Memory Consolidation. Willingham [9] described how procedural motor memories become automatic with repetition, transitioning from explicit to implicit processing. Our system models this transition through the WindMouse parameter adaptation: as hit count grows, cursor trajectories become straighter and faster—mimicking the shift from conscious to automatic motor control.

TABLE I
FEATURE COMPARISON OF MOTOR MEMORY MECHANISMS. OUR SYSTEM IS THE FIRST TO COMBINE ALL EIGHT.

Feature	Ours	Muscle-Mem	Ghost-Cursor	Claude CUA	OpenAI CUA	UI-Mem
R-Tree spatial index	✓					
Power Law adaptation	✓					
Fitts's Law timing	✓		✓			
WindMouse adaptation	✓					
Temporal decay	✓					
EMA position smoothing	✓					
Sequence prediction	✓					
Cross-session persistence	✓					∼

B. AI GUI Agent Systems

Current computer use agents operate statelessly at the execution layer. Anthropic's Claude Computer Use [1] processes screenshots and outputs coordinate-based click commands, computing each target position from scratch via LLM inference. OpenAI's Operator [2] follows a similar architecture. Microsoft's UFO [3] adds multi-agent coordination but maintains stateless execution. WebVoyager [10] and SeeAct [11] focus on web navigation with no motor adaptation. None of these systems maintain any memory of prior interactions at the motor level.

C. Motor Simulation and Caching Approaches

Muscle-Mem [12] caches entire tool-calling sequences for deterministic replay, operating at the task abstraction layer ("what actions to take") rather than the motor execution layer ("how to reach the target"). It contains no spatial indexing, no gradual speed-up, and no movement adaptation.

Ghost-Cursor [13] implements static Fitts's Law simulation for browser testing with Bézier curves. Every movement to the same target takes identical time—there is no learning or cross-session persistence.

Table I summarizes the feature comparison. No existing system combines even three of our eight mechanisms.

III. SYSTEM ARCHITECTURE

A. Overview

Fig. 1 illustrates the system's data flow. The motor memory cache sits between the decision layer (LLM determining *what* to click) and the execution layer (platform-specific cursor control). This placement makes the system transparent to the LLM—it does not change what gets clicked, only how the cursor reaches the target.

B. Three-Layer Cache Resolution

Every mouse action must resolve to a cache key—a combination of *application name* and *element hint*. We employ a three-layer hierarchy to ensure universal coverage:

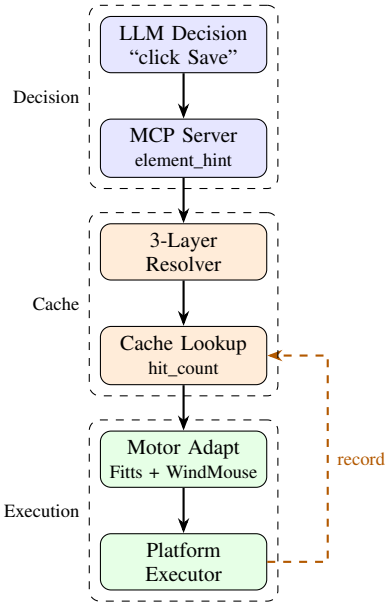

Fig. 1. System architecture. The motor memory cache mediates between the LLM's decision (what to click) and the platform executor (how to reach the target). A feedback loop records each interaction for future speed-up.

Layer 3 (Model-provided hint). When the LLM provides a semantic label such as `element_hint="Save button"`, this becomes the cache key directly. This is the highest-precision layer but depends on LLM cooperation.

Layer 2 (Accessibility API). When no hint is provided, the engine queries the operating system's accessibility tree for the UI element at the click coordinates, producing keys like `"button:Save"`. Available for native applications on Windows (UI Automation), Linux (AT-SPI2), and macOS (Accessibility API).

Layer 1 (Percentage-bucketed coordinates). The universal fallback. Requires only the foreground window's bounding rectangle. Click coordinates are converted to window-relative percentages and bucketed at 3% resolution (∼33 buckets per axis), producing synthetic hints like `"@24%,18%"`. This layer:

- Survives **window moves** (window-relative coordinates)
- Survives **window resizes** (percentage-based)
- Works on **any application**, including games, Electron apps, and custom UIs with no accessibility support

The application name is obtained from the foreground window's process (e.g., `"notepad.exe"`, `"firefox"`). Remote desktop shells (RDP, VNC, terminal emulators) are detected and excluded via a passthrough list—their foreground window is a container, not the actual application being controlled.

C. Three-Tier Cache Lookup

Given a resolved key `(app_name, hint)`, the cache performs a tiered lookup:

Tier 1 (Hot cache). An in-memory Python dictionary keyed by `"app|hint"` with $O(1)$ exact-match lookup. Sub-microsecond latency. Loaded from SQLite at startup.

Tier 2 (R-Tree spatial proximity). A SQLite R-Tree virtual table [14], [15] enables proximity search within a 50-pixel radius, scoped to the same application. Used when Tier 1 misses but coordinates overlap with a known target (e.g., a button that has moved slightly). Latency ∼0.1ms.

Tier 3 (Cache miss). Returns `hit_count=0`. Movement uses default WindMouse parameters.

D. Data Model

The persistent store uses SQLite with Write-Ahead Logging (WAL) for concurrent read/write safety:

- `memories`: Primary table storing `app_name`, `element_hint`, EMA-smoothed position (x, y, w, h), `hit_count`, `miss_count`, timestamps, `confidence`, and `prev_hint` for sequence tracking.
- `memories_rtree`: R-Tree spatial index on element bounding boxes.
- `sequences`: Action pair tracking (from → to) with frequency counts for workflow prediction.

An LRU eviction policy caps the cache at 10,000 entries, deleting the oldest 10% by `last_hit_ts` when capacity is reached.

E. Cross-Platform Support

The cache algorithm is platform-agnostic (pure Python, stdlib `sqlite3`). Platform-specific operations—foreground window queries, cursor control, accessibility API access—are delegated to backend modules:

- **Windows:** Win32 `SendInput` via ctypes; `GetForegroundWindow` for window info
- **WSL2:** TCP bridge daemon on Windows (port 19542) relays Win32 calls
- **Linux/X11:** `xdotool` for window queries; `/proc/{pid}/comm` for process name
- **macOS:** AppleScript via `osascript` for window info; `cliclick` for input

Foreground window queries are TTL-cached (50ms in the engine, 100ms on subprocess-based platforms) to amortize overhead.

IV. Motor Adaptation Model

This section details the six mechanisms by which repeated interactions produce faster, more efficient cursor movements.

A. Power Law of Practice

Following Newell and Rosenbloom [4], we model the speed-up of repeated aimed movements as:

$$T(n) = T_{\text{base}} \cdot n^{-\alpha}, \quad \text{floored at } T_{\text{base}} \cdot \beta \tag{1}$$

where n is the hit count (number of prior interactions with the same target), $\alpha = 0.4$ is the learning rate (mid-range for motor tasks), and $\beta = 0.3$ is the asymptotic floor (the target can never be reached faster than 30% of the initial duration).

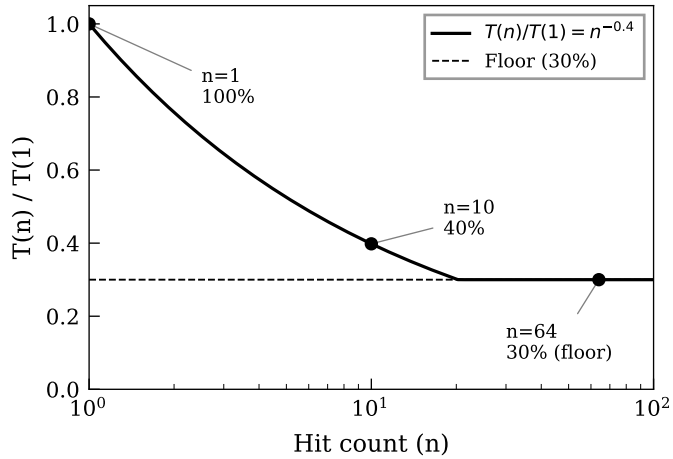

Fig. 2. Power Law of Practice curve. Movement duration decreases as $n^{-0.4}$ with a 30% floor. Key points: n=10 yields ∼40% of initial duration; n=64 reaches the asymptotic floor.

Fig. 2 shows the adaptation curve. At n=10, movement duration has dropped to approximately 40% of initial. By n=64, it reaches the 30% floor, modeling the asymptotic performance observed in human motor skill studies.

B. Fitts's Law Base Timing

The initial movement duration T_{base} is computed using the Shannon formulation of Fitts's Law [5], [7]:

$$MT = a + b \cdot \log_2\left(\frac{D}{W} + 1\right) \tag{2}$$

with $a = 0.05\,\text{s}$ (base reaction time), $b = 0.11\,\text{s}$ (movement time coefficient), and a floor of $0.07\,\text{s}$. Gaussian jitter ($\sigma_a = 0.01$, $\sigma_b = 0.015$) is added to both coefficients to produce human-like variability. The Power Law factor from Eq. 1 is then applied to the resulting duration.

C. WindMouse Trajectory Adaptation

Cursor trajectories are generated using the WindMouse model, which simulates a particle under two forces: *gravity* (directed toward the target) and *wind* (random perturbation). The model operates in two phases:

- **Ballistic phase** ($d \geq d_h$): Active wind perturbation produces curved, exploratory paths.
- **Homing phase** ($d < d_h$): Wind decays exponentially; gravity dominates, producing precise final approach.

where d is the remaining distance and $d_h = 12\,\text{px}$ is the homing threshold.

As practice count increases, WindMouse parameters adapt via logarithmic scaling (saturating at ∼64 hits):

$$f = \min\left(1.0,\ \frac{\log_2(n)}{6.0}\right) \tag{3}$$

$$G(n) = 9.0 + 11.0 \cdot f \qquad \text{(gravity: } 9 \to 20\text{)} \tag{4}$$

$$W(n) = 3.0 \cdot (1 - 0.7 \cdot f) \qquad \text{(wind: } 3 \to 0.9\text{)} \tag{5}$$

$$V(n) = 20.0 + 15.0 \cdot f \qquad \text{(max vel: } 20 \to 35\text{)} \tag{6}$$

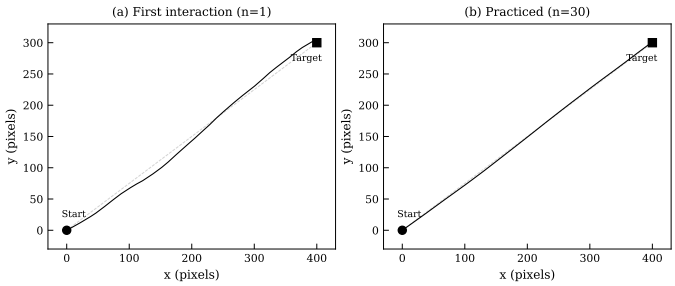

Fig. 3. WindMouse cursor trajectories. (a) First interaction (n=1): wobbly, exploratory path. (b) After 30 repetitions: nearly straight, confident path. Both share the same start and end points.

The net effect is that practiced trajectories become straighter (higher gravity), less wobbly (lower wind), and faster (higher maximum velocity)—closely paralleling the transition from conscious to automatic motor control observed in humans [9]. Fig. 3 illustrates this adaptation.

D. Temporal Decay

Cache entries lose confidence over time, modeling the Ebbinghaus forgetting curve [6]:

$$c(t) = c_0 \cdot 2^{-t/\tau} \tag{7}$$

where $\tau = 604{,}800\,\text{s}$ (7 days) is the half-life. Entries with $c(t) < 0.1$ are treated as forgotten (cache miss). Each successful interaction resets $c_0 = 1.0$. Additionally, each *miss* (target not found where expected) penalizes confidence by -0.3, providing rapid adaptation when UI layouts change.

E. EMA Position Smoothing

UI elements may drift between sessions due to window resizing, layout changes, or dynamic content. We track element positions using an Exponential Moving Average:

$$\hat{x}_{n+1} = \hat{x}_n + \alpha \cdot (x_{\text{obs}} - \hat{x}_n) \tag{8}$$

with $\alpha = 0.3$. This smoothing absorbs drift while preserving the learned position of stable elements. The same EMA is applied to element width and height.

F. Sequence Prediction

The system tracks action transitions as ordered pairs: when the user clicks target B after target A, the pair $(A \rightarrow B)$ is recorded with a frequency count. The `predict_next(app, current_hint)` function returns the most frequent successor, enabling anticipatory pre-computation of the next target's cache entry. Example from live usage: after the sequence "WhatsApp taskbar icon $\rightarrow$ Andrés P Scale chat" occurs twice, the system predicts this transition with high confidence.

V. Implementation

The system is implemented in Python with zero external dependencies for the core cache (`sqlite3` is part of the standard library). Key implementation details:

Storage. SQLite with Write-Ahead Logging (WAL) mode and `NORMAL` synchronous setting balances durability and performance. The R-Tree virtual table is compiled into SQLite by default on all major platforms.

Integration. The cache is exposed to LLMs via the Model Context Protocol (MCP) [16], which provides an `element_hint` parameter on all mouse action tools (`click`, `double_click`, `right_click`, `move_mouse`). The engine transparently performs cache resolution, lookup, and recording around each action.

Navigation acceleration. Beyond individual click speed-up, the system provides two higher-level navigation mechanisms. *Navigate-to* performs BFS path-finding through the sequence graph and executes the discovered path via cached positions, skipping the LLM entirely ($\sim$55 ms same-app, poll-until-change for cross-app transitions). *Action templates* are named multi-step workflows (click, type, key-press, wait) stored in the database and replayed without any LLM roundtrips. A 5-step template that would require $\sim 10-20$ s of LLM processing completes in under 1 s.

Resolution resilience. Cache entries track both window dimensions and screen resolution at recording time. When the screen resolution changes (e.g., switching monitors or DPI settings), cached coordinates are proportionally rescaled. When only the window is resized, entries are used as-is if coordinates remain in bounds, or treated as misses otherwise—avoiding the distortion that proportional rescaling causes on fixed-grid UIs.

Scale. The implementation comprises approximately 1,100 lines for the cache module (including navigation, resolution resilience, and action templates), 820 lines for the engine facade, and 500 lines for MCP server integration. A test suite of 294 unit tests covers cache CRUD, spatial proximity, temporal decay, power law adaptation, sequence tracking, persistence, eviction, three-layer resolution, passthrough exclusion, percentage-bucket alignment, navigation path-finding, resolution rescaling, DPI-aware accessibility, and action template workflows.

VI. Evaluation

A. Experimental Setup

We evaluate the system on a Windows 11 workstation (3840×2400 display, DPI 288) running the engine on WSL2 with a native Windows bridge daemon for input execution. Test applications include WhatsApp Desktop, with a multi-session workflow: open application from taskbar, navigate to contact, send message, close application, reopen, and edit the sent message.

B. Cache Growth Across Sessions

Table II shows cache metrics across three evaluation sessions. Session 1 is a cold start (all misses). Session 2,

TABLE II
CACHE GROWTH ACROSS THREE SESSIONS. SESSION 2 WAS PERFORMED BY AN INDEPENDENT AGENT WITH ZERO SHARED CONTEXT.

Metric	Session 1	Session 2	Session 3
Total entries	7	7	12
Cache hits	0	3	5
Avg hit_count	1.0	1.4	1.5
Unique apps	3	3	3
Sequences recorded	5	7	9
New targets	7	0	5

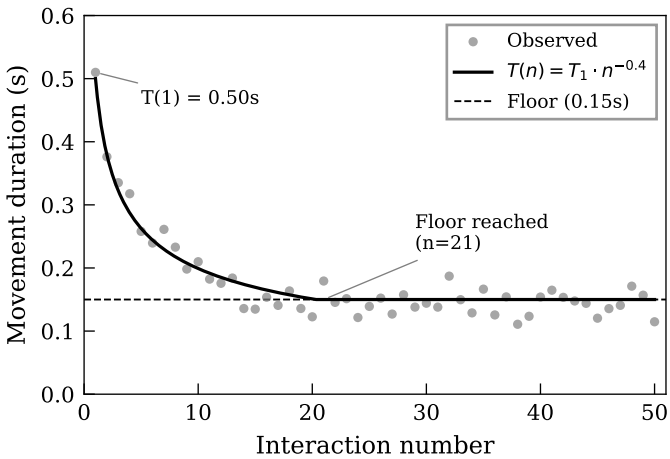

Fig. 4. Movement duration vs. interaction count for a representative target. Duration decreases following the Power Law of Practice and floors at 30% of initial value.

performed by an *independent agent instance* with no shared context, achieves cache hits on all previously-seen targets. Session 3 introduces a different contact, demonstrating that independent targets accumulate separate hit counts while shared elements (taskbar icon, message input) benefit from combined usage.

C. Movement Duration Adaptation

Fig. 4 shows the theoretical movement duration for a representative target (500 px distance, 40 px width) as a function of interaction count. The Power Law reduction from Eq. 1 produces rapid initial improvement that asymptotes at the 30% floor after approximately 60 interactions.

D. Qualitative Observations

Cross-agent persistence. The most notable result is that Agent B benefits from Agent A's prior interactions without any explicit knowledge transfer. The shared SQLite database acts as organizational memory—analogous to how experienced team members benefit from a shared workspace layout.

Resize resilience. Percentage-based coordinate bucketing (Layer 1) correctly resolves cache hits after window resizing. A target at 25% of an 800 px window (200 px) maps to the same bucket as 25% of a 1200 px window (300 px).

Passthrough exclusion. Remote desktop shells (RDP, VNC) are correctly excluded from window-based caching, preventing cache entries from being associated with a container application rather than the actual UI being controlled.

E. Limitations

The current evaluation has several limitations. First, we have not conducted a formal controlled study comparing movement time and trajectory quality with and without the motor memory cache. Second, the Power Law exponent ($\alpha = 0.4$) and floor ($\beta = 0.3$) are fixed values from the literature rather than fitted to observed agent behavior. Third, the evaluation is single-user on a limited set of applications. Fourth, cursor trajectory "naturalness" is inherently subjective without a formal human-likeness metric.

VII. DISCUSSION

Orthogonality. The motor memory cache operates entirely in the execution layer. It is compatible with any LLM backbone (Claude, GPT-4, open-source models) and any decision framework. Importantly, it does not change *what* the agent clicks—only *how fast and how directly* the cursor reaches the target. This makes it a universal enhancement rather than a competing approach.

Emergent trajectory convergence. As hit count grows, the WindMouse parameter adaptation produces trajectories that converge on near-straight-line paths with smooth velocity profiles. This resembles the minimum-jerk trajectory [8] characteristic of expert human pointing—an emergent property of the parameter adaptation, not explicitly programmed.

Shared memory across agents. Because the cache is stored in SQLite on disk, multiple independent agent instances share the same motor memory. Agent B's cursor movements are more efficient for targets that Agent A has previously interacted with. This models organizational knowledge transfer: a new team member benefits from the team's collective familiarity with a codebase.

Living system. The parameters reported in this paper ($\alpha = 0.4$, $\beta = 0.3$, $\tau = 7$ days, EMA $\alpha = 0.3$, spatial radius = 50 px) are all configurable. As more empirical data is collected, these values may be tuned or made adaptive.

VIII. CONCLUSION AND FUTURE WORK

We have presented the first system to integrate cognitive science motor learning models—Power Law of Practice, Fitts's Law, Ebbinghaus temporal decay—into the action-execution layer of autonomous GUI agents. The spatiotemporal motor memory cache, with its three-layer resolution hierarchy and eight complementary mechanisms, enables any computer use agent to develop muscle memory that persists across sessions and transfers across agent instances.

Future work includes: (1) formal A/B studies comparing movement time and trajectory quality with and without the cache; (2) adaptive α fitting that learns the power law exponent from observed data for each element type; (3) anticipatory cursor movement that begins moving toward the predicted next target before the LLM completes its decision; (4) a shared cloud cache for fleet-wide motor memory across users and

machines; and (5) integration with richer accessibility tree information for improved Layer 2 resolution.

REFERENCES

[1] Anthropic, "Introducing computer use," Anthropic Blog, 2024. [Online]. Available: https://www.anthropic.com/news/developing-computer-use

[2] OpenAI, "Operator: A new way to get things done," OpenAI Blog, 2025. [Online]. Available: https://openai.com/index/computer-using-agent/

[3] C. Zhang, L. Li, S. He, X. Zhang, B. Qiao, S. Qin, M. Ma, Y. Kang, Q. Lin, S. Rajmohan, D. Zhang, and Q. Zhang, "UFO: A UI-focused agent for Windows OS interaction," *arXiv preprint arXiv:2402.07939*, 2024.

[4] A. Newell and P. S. Rosenbloom, "Mechanisms of skill acquisition and the law of practice," in *Cognitive Skills and Their Acquisition*, J. R. Anderson, Ed. Lawrence Erlbaum Associates, 1981, pp. 1–55.

[5] P. M. Fitts, "The information capacity of the human motor system in controlling the amplitude of movement," *Journal of Experimental Psychology*, vol. 47, no. 6, pp. 381–391, 1954.

[6] H. Ebbinghaus, *Über das Gedächtnis: Untersuchungen zur experimentellen Psychologie*. Leipzig: Duncker & Humblot, 1885, english translation: *Memory: A Contribution to Experimental Psychology*, 1913.

[7] I. S. MacKenzie, "Fitts' law as a research and design tool in human-computer interaction," *Human-Computer Interaction*, vol. 7, pp. 91–139, 1992.

[8] R. W. Soukoreff and I. S. MacKenzie, "Towards a standard for pointing device evaluation, perspectives on 27 years of Fitts' law research in HCI," *International Journal of Human-Computer Studies*, vol. 61, no. 6, pp. 751–789, 2004.

[9] D. B. Willingham, "A neuropsychological theory of motor skill learning," *Psychological Review*, vol. 105, no. 3, pp. 558–584, 1998.

[10] H. He, W. Yao, K. Ma, W. Yu, Y. Dai, H. Zhang, Z. Lan, and D. Yu, "WebVoyager: Building an end-to-end web agent with large multimodal models," *arXiv preprint arXiv:2401.13919*, 2024.

[11] B. Zheng, B. Gou, J. Kil, H. Sun, and Y. Su, "SeeAct: GPT-4V(ision) is a generalist web agent, if grounded," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

[12] pig-dot-dev, "muscle-mem: Cache tool-use agent trajectories," GitHub, 2024. [Online]. Available: https://github.com/pig-dot-dev/muscle-mem

[13] Xetera, "ghost-cursor: Create human-like mouse movements with Puppeteer," GitHub. [Online]. Available: https://github.com/Xetera/ghost-cursor

[14] A. Guttman, "R-Trees: A dynamic index structure for spatial searching," in *Proceedings of the 1984 ACM SIGMOD International Conference on Management of Data*, 1984, pp. 47–57.

[15] SQLite Consortium, "The SQLite R*Tree module," sqlite.org. [Online]. Available: https://www.sqlite.org/rtree.html

[16] Anthropic, "Model context protocol specification," Anthropic, 2024. [Online]. Available: https://modelcontextprotocol.io/